INCH-POUND

MIL-R-43837B(GL)
12 January 1994
SUPERSEDING
MIL-R-43837A(GL)
31 March 1983

MILITARY SPECIFICATION

RELEASE, CARGO PARACHUTE, AIRDROOP, M-1

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cargo parachute release assembly consisting of a release machanism, upper and lower suspension link assemblies and cargo parachute connectors. The cargo patachute connections and retaining clamp assemblies may be procured separately (see 6.2).

2. APPLICATION DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-H-6875 - Heat Treatment of Steel process For

(Beneficial comments recommendations, additions, deletions) and any pertinent data which may be used in improving this documnent should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for ispection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-147	- Palletized Unit Loads
MIL-STD-2073-1	- DoD Materiel procedures for Deployment and Application of Packaging Reguirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Goverment documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-894	- Timer Delay Assembly (Release, Cargo Parachute, M1 & M2)
11-1-1487	- Release Assembly, Cargo Parachute, Airdrop, M-1
11-1-1489	- Plate, Back
11-1-1491	- Toggle
11-1-1500	- Plate, Front

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering Center, ATTN: SATNC-MII, Hatick, MA 01760-5017.)

2.2 Non-Government publications. The follwing documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, teh issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Application for copies should be addressed to the America National Standards Institute, Inc., 11 West 42nd Street, New York NY 10036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard practice for Commercial Packaging

(Appliation for copies should be addressed to the American Society for resting and Materials, 1916 Race Street, Philadelphia PA 19103-1187.)

(Non-Government standards and other pulications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components used in the manufacture of the M-1 cargo parachute release shall conform in all respects to the requirements specified or shown on Drawing 11-1-1487-2, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

3.3 Construction and supplementary process. The construction and supplementary process used in the manufacture of the cargo parachute releases shall conform to the requirements specified or shown on Drawing 11-1-1487-2, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

3.3.1 Surface finish. American National Standard Institute (ANSI) Standard No. B46.1, shall apply for the interpretation of surface roughness values specified on the applicable drawings.

3.4 Performance.

3.4.1 Cargo parachute connector proofload. The cargo parachute connector shall show no visual evidence of failure (fracture or permanent deformation) when subjected to a tensile proofload of not less than 25,000 pounds for not less than 15 seconds when tested as specified in 4.4.5.

3.4.2 Release assembly proofload. The release assembly shall show no visual evidence of failure (fracture or permanent deformation) when subjected to a tensile proofload of not less than 52,000 pounds for not less than 15 seconds when tested as specified in 4.4.5.

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3.4.3 Release assembly unlocking. A completion of the timer unwinding cycle, the timer delay with attached toggle locking slides shall drop to the unlocked bottom position with no evidence of interference when tested as specified in 4.4.5.

3.4.4 Retaining clamp assembly proofload. The retaining clamp assembly shall show no visual evidence of failure or permanent deformation at when subject to a tensile proofload of not less than 52,000 pounds for not less than 15 seconds, when tested as specified in 4.4.5.

3.5 Identificaiton marking. Identification marking of release assemblies and all components procured separately shall be in accordance with MIL-STD-130 and with the legend and location as shown on applicable drawings.

3.6 Government furnished property (GFT). The following items, when required for proofload testing of the release assembly or separately procured cargo parachute connectors or retaining clamp assembly, may be obtained from the procuring activity on a loan basis (see 6.4).

Release Assembly Cargo Parachute, Airdrop M-1 or M-2
Part No. 11-1 -1487-1 or 11-1-1487-2
11-1-555-1 or 11-1-565-2

Timer Delay Assembly; Releases, Cargo, Parachute, M-1 and M-2
Part No. 11-1-894-1

3.7 Workmanship. The release assembly and its components shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspecion requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspecition requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to precribed requirements.

4.1.1 Responsibility for compliance. All item shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specificaiton shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Goverment for acceptance comply with

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all requirements of trhe contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain informance to requirements, however, this does not authorize submission of known defective material. either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufactureing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 5.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of reference documents unless otherwise excluded, amended, modified, or quaified in this specification or applicable purchase document.

4.4.2 In-process inspections. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process heat treatment inspection. Heat treated components shall be inspected to determine compliance with requirements specified on the applicable drawings. The provisions of MIL-H-6875 applicable to routine quality control shall apply, except that the inspection level shall be S-2. The finding of any defect shall be cause for rejection of the lot.

4.4.2.2 In-process dimensional examination. Prior to assembly, but after heat treatment and application of protective finish; the item Listed below shall be examined for conformance to the dimensions indicated. The lot size shall be expressed in units of items of appicable type. The sample unit shall be S-3. The finding of any defect shall be cause for rejection of the lot.

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ITEM	Drawing	Dimension
Toggle	11-1-1491	.344 and .500
Rent plate	11-1-1500	.348 and .500
Back plate	11-1-1489	.348 and .500

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table 1. The lot size shall be expressed in units of complete release assemblies, cargo parachute connectors, or retaining clamps as applicable. The sample unit shall be one complete release assembly, four cargo parachute connectors, or one retaining clap as applicable. The inspection level shall be II. The finding of any defect shall be cause for rejection of the lot.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Omitted or not type specified	101	
	Coating powdery, coarse grained, mottled, not adherent, evidence of corrosion	102	
	Supplementary treatment omitted, not type specified, tacky	103	
	Color not as specified		201
	Dirt, oil, grease or other foreign matter		202
Design	Any departure from design specified on drawings	104	
Construction and workmanship	surf ace roughness value in excess of rating specified on drawing	105	
	Any component missing, cracked, visually malformed, misaligned, damaged	106	
	sharp edges , burnes or metal slivers	107	
	Components incorrectly assembled	108	
	Components loose	109	
	Movement of parts inhibited	110	
	Threaded sections damaged	111	
Identification	Missing, incomplete, illegible, improper location, or not as specified		203

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4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of complete release assemblies, cargo parachute coonnectors, or retaining clamps as applicable. The sample unit shall be one complete release assembly, four cargo parachute connectors, or one retaining clamp as applicable. The inspection level shall be S-3. The finding of any defect shall be cause for rejection of the lot.

4.4.5 End item testing. Cargo parachute connectors, release assemblies (proofload and unlocking), and retaining clamp assmblies shall be tested as specified in 4.5.1, 4.5.2 and 4.5.3 for compliance with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4 respectively. The lot size shall be no more than 50 completely assembled cargo parachute connectors, complete release assemblies or retaining clamp assemblies. The sample unit shall be one completely assembled cargo parachute connector, one complete release assembly or one retaining clamp assembly. Cargo parachute connectors or retaining clamp assemblies showing any visual evidence of failure (fracture or permanent deformation) shall be cause for rejection of the lot. Release assemblies showing any visual evidence of failure (fracture or permanent deforamtion) of any component of the release assembly, exclusive of the cargo parachute connectors and time delay assembly, or failure of the delay assembly and attached lock slides to drop freely without interference to the unlocked bottom position shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be oneshipping container fully packaged. The inspeciton level shall be S-2. The findingof any defect shall be cause for rejection of the lot.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of inproper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.4.7 Palletization examination. The fully packaged and Palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1. The finding of any defect shall be cause for rejection of the lot.

Examine	Defect
Finished diomensions	Lenght, width, or height exceeds specified specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Cargo parachute connector proofload testing. The cargo parachute connector to be testd shall be installed in a contractor furnished test fixture. The test fixture shall be attached to a direct reading and calibrated tensile testing device capable of imposing a 25,000 pound tensile force on the fixture and test its. The test fixture shall be constructed essentially in accordance with figure 1 and designed with a minimum safety factor of 2.0. The connector sleeve may be removed to facilitate attachment of the connector body to the test fixture. subject to Government approval, a mechanism may be developed to restrain the connector arms in the same manner in which they are normally attached to the release assembly in lieu of the release and time delay assemblies shown on figure 1. A proofload of 25,000 pounds shall be applied to the connector at a rate of not less than 3,000 pounds/second nor greater than 6,000 pounds/second. The connector shall withstand the tensile load for 15 seconds, after which the load shall be reduced to zero. The proofload of 25,000 pounds shall be immediately reapplied, held for 15 seconds, and then reduced to zero.

4.5.2 Release assembly proofload and unlocking testing. The release assembly shall be assembled with four cargo parachute connectors and with a timer delay assembly (GFP) installed in the locked position. The locked position requires raising the timer delay assembly with the attached toggle lock slides, turning the timer stem approximately 90 degrees clockwise to engage the timer keys with the release cavity slots, and inserting the arming wire through the slot provided and through the timer stem. The complete assembly shall then be mounted in a contractor furnished test fixture. The text fixture shall be attached to a calibrated tensile testing machine capable of imposing a 52,000 pound tensile force on the text fixture and release assembly. The test fixture shall be fabricated in accordance with figure 2 and designed with a minimum safety factor of 2.0. The fixture shall

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provide that the total applied tensile force be equally districted among the four cargo parachute connectors and also divided (equally between the two liner link assemblies. The cargo parachute connector sleeves may be remove to facilitate attachment of the connector bodies to the test fixture. the lower link assemblies shall be connected so that only one branch of each link assembly will be subjected to the tensile load. The proofload of 52,000 pounds shall be applied to the release assembly at a rate of not less than 5,000 pounds per second, nor greater than 10,000 pounds per second. The release assembly shall withstand the tensile loas for 15 seconds, after which the load shall be reduced to zero. The 52,000 pound proofload shall be immediately reapplied, held for 15 seconds, and then reduced to zero. After completion of the second proofload application, a tensile load not to exceed 100 pounds shall be applied to the release assembly. The release assembly shall then be activated by removing the arming wire. After the timing cycle, the delay assembly with attached toggle lock slides shall drop within the release body to the unlocked (bottom) position. The tensie load shall then be reduced to zero and the release assembly disassembled for inspection.

4.5.3 Retaining clam assembly proofload testing. The retaining clamp shall be assembled in a parachute release with four cargo parachute connectors and with a timer delay assembly installed in the locked position. The locked position prevents the parachute connectors from, separating from the release. The locked position requires raising the timer delay assembly with the attached toggle lock slides, turning the timer stem approximately 90 degrees clockwise to engage the timer keys with the release cavity slots and inserting the arming wire through the slot provided and through the timer stem. The complete assembly shall then be mounted in a contractor furnished test fixture. The test fixture shall be attached to a calibrated tensile testing machine capable of imposing a 52,000 pound tensile force on the test fixture and release assembly. The test fixture shall be fabricated in accordance with figure 2 and designed with a minimum safety factor of 2.0. The fixture shall provide that the total applied tensile force be equally distributed among the four cargo parachute connectors and also divided equally between the two lower link assemblies. The cargo parachute connector sleeves may be removed to facilitate attachment of the connector bodies to the test fixture. The proofload of 52,000 pounds shall be applied to the release assembly at a rate of not less than 5,000 pounds per second, nor greater than 10,000 pounds per second. The release assembly shall withstand the tensile load for 15 seconds, after which the load shall be reduced to zero. The 52,000 pound proofload shall be immediately reapplied, held for 15 seconds, and then reduced to zero. The parachute release shall be disassembled and the retaining clamp assembly examined.

4.5.4 Procurement of cargo parachute connectors as a replacement unit. The cargo parachute connectors when procured as a replacement or spare part (see 6.2) shall be tested for the requirements of 3.4.1 and conformance to 4.4.5. A complete and operable release assembly and timer delay assembly will be provided (see 3.6) for conduct of the proofload test, if required.

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4.5.5 Procurement of retaining clamps as a replacement unit. The retaining clams when procured as a replacement or spare part (see 6.2) shall be tested for the requirements of 3.4.4 and conformance to 4.4.5. A complete and operable release assembly and timer delay assembly will be provided (see 5.6) for conduct of the proofload test, if required.

5. PACKAGING

5.1 Preservation and packing. Levels of preservation and packing shall be in acordance with the procurement order/contract or in accordance with MIL-STD-2073-1 and ASTM-D-3951 (see 6.2).

5.2 Palletization. When specified (see 6.2), palletization shall be in accordance with the procurement order/contract or MIL-STM-147.

5.3 Marking. Marking shall be in accordance with the procurement order/contract or MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Release, cargo Parachute, Airdrop M-1, covered by this document is used to automatically the parachute from the load at the moment of ground impact. This prevents the parachute from dragging or overturning the load. This release intended for loads having a gross rigged weight of up to 15,000 pounds.

6.2 Acquisition requirements. Acquistion documents must specify the following:

a. Title, number, and date of this specification.
b. When a separate procurement of cargo parachute connetors and retaining clamp assemblies are required (see 1.1.).
c. Issue od DODISS to be cited in the solicitation, and if required the specific issue of individual documetns referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1).
f. When palletization is required (see 5.2).

6.3 First article. When a first article is required,it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-3 or 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished.

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The contracting officer should also include specific instructions in acquisition documents regarding arrangments for selection, inspection, and approval of the first article.

6.4 Government furnished property (GFP). For access to government furnished property (see 3.6) contact the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key work) listing.

Parachute mechanism

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army-GL

Review activity:

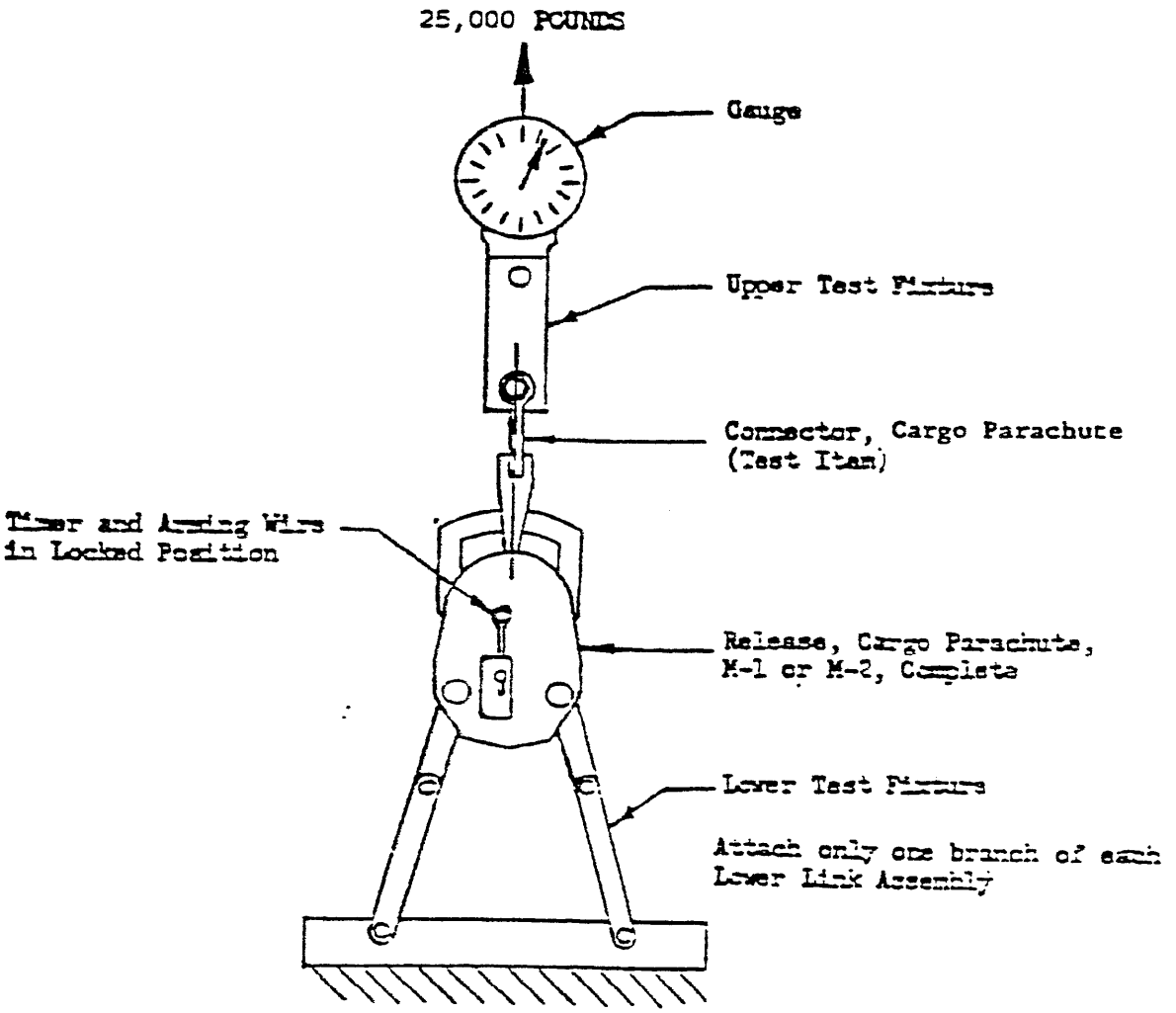
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Preparing activity:

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(Project 1670-A808)

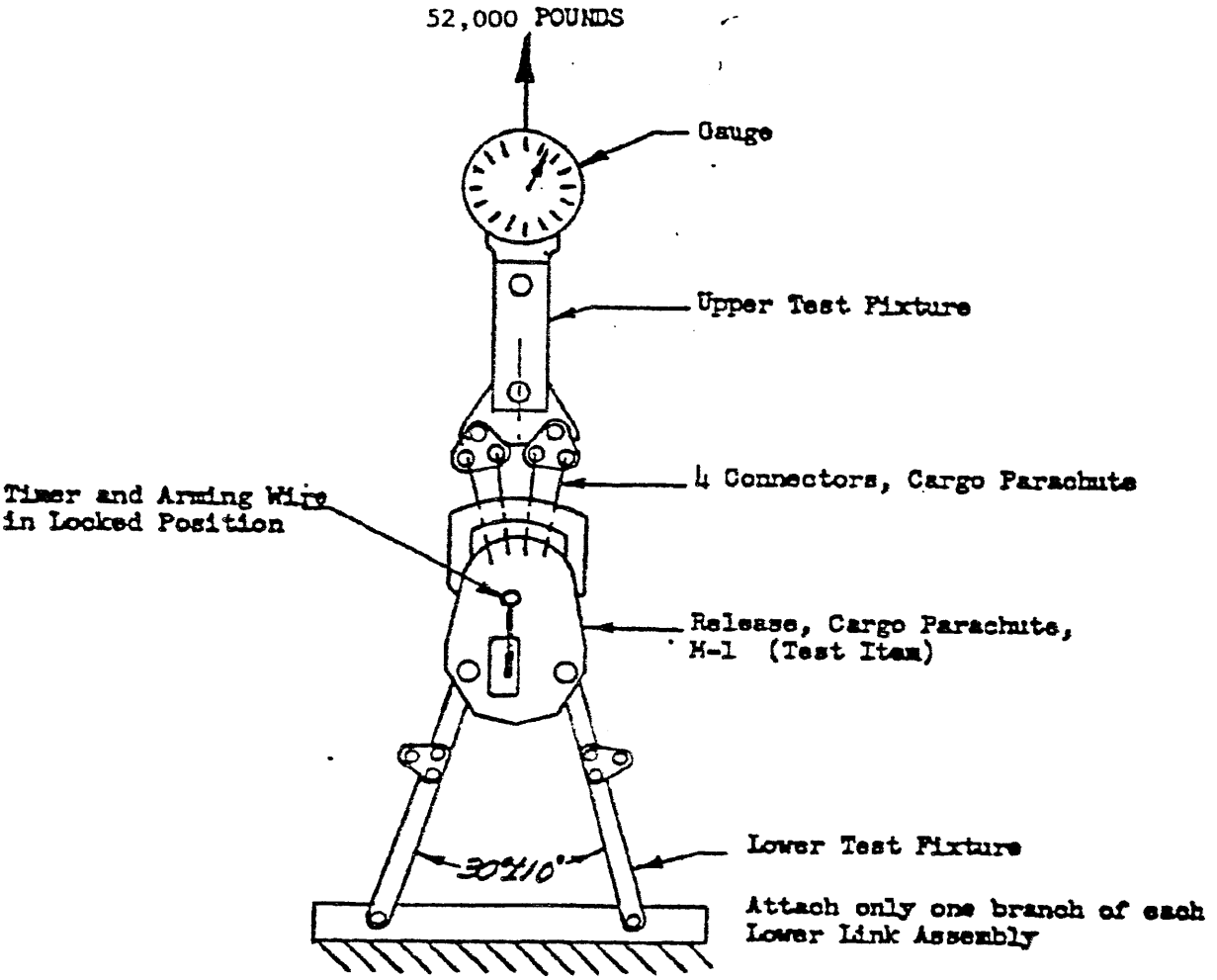
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PROOF LOAD, CONNECTOR, CARGO PARACHUTE

FIGURE 1

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PROOF LOAD TEST M-7

FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-43837B(GL)	94/01/12

3. DOCUMENT TITLE
RELEASE, CARGO PARACHUTE, AIRDROP, M-1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-5235	256-5235
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-U C Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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